

Software Requirements Specification

for

DAV - Voice-Assisted Design in FreeCAD

Version 0.1

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Revision History

Name	Date	Reason for Changes	Version
Team	15/05/2026	Initial version	0.1

1. Introduction

1.1. Purpose

This document specifies the software requirements for the DAV product, aimed at enabling students and people with motor, visual, or cognitive disabilities to create and modify designs, drawings, or 3D models in FreeCAD through voice commands (for example: create a box, define dimensions, color, or diameter), complementing traditional interaction whenever possible.

The document identifies the product, its revision (0.1), and the scope covered by this specification: voice interface, command interpretation layer, integration with FreeCAD, and associated quality requirements.

This document does not replace the end-user manual or the detailed technical implementation documentation.

1.2. Document Conventions

- Functional requirements are expressed using the term “shall” and use identifiers in the format REQ-x.y.
- Non-functional requirements use the prefix NF-x.y.

Priorities are classified as follows:

Priority	Meaning
Must	Essential for the MVP. Important but non-blocking features for the initial release.
Should	Desirable if time permits
Could	Future functionality. Optional features or quality-of-life improvements to be evaluated later.

- Command examples are shown in quotation marks.
 - Technical terms (ASR, API, NLU) are defined upon their first appearance.
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1.3. Intended Audience and Reading Suggestions

Reader Type	Document Usage
Teachers / Evaluators	Scope and objectives
Developers	Interfaces and requirements
UX / Accessibility Design	Feedback and error handling
Management / Project Management	Scope and planning

Suggested Reading Order:

1 → 2 → 3 → 5 → 4 → 6

1.4. Product Scope

The DAV software is intended to provide an interaction model that translates spoken instructions into actions performed on FreeCAD documents:

- Basic geometry creation
- Dimension modification
- Visual configuration
- Document operations
- Clear user feedback

Benefits

- Reduces dependence on mouse interaction
- Promotes educational inclusion
- Maintains compatibility with FreeCAD

Objectives

1. Clear and repeatable voice commands
2. Confirmation of critical actions
3. Compatibility with keyboard and mouse input
4. Predictable error handling

Out of Scope (MVP)

- Full replacement of FreeCAD
 - Advanced FEA simulation
 - Voice-based sketch creation
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1.5. References

- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications
 - IEEE 830 Template
 - FreeCAD Documentation and Python API
 - ISO/IEC 25010
 - WCAG 2.x
 - Project Description
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2. Overall Description

2.1. Product Perspective

DAV is a new product that operates as a voice assistance layer on top of FreeCAD.

It does not modify or replace FreeCAD functionality, but instead works as an extension or complementary component.

The project originates from educational and accessibility needs.

2.2. Product Features

Voice Input

- Interaction via spoken commands using ASR and NLU recognition technologies.

Basic Geometry

- Ability to generate 3D bodies (boxes, cylinders, spheres) using voice.

Parameters

- Configuration and modification of part dimensions (height, width, depth, radius, diameter).

Appearance

- Aesthetic customization of models through colors and materials.

Document and Security

- Project file management (new, open, save) with confirmation mechanisms for destructive actions.

Safety

- Confirmation for destructive actions

Compatibility

- Ability to operate simultaneously with traditional peripherals such as keyboard and mouse.

2.3. User Classes and Characteristics

Class	Characteristics	Importance
Student with motor disabilities	Prioritizes voice interaction	High
General student	Complementary use	Medium
Teacher	Requires stability	Medium
Technical administrator	Configuration tasks	Low

2.4 Operating Environment

Hardware

- FreeCAD compatible PC
- Microphone
- Headphones or speakers

Software

- FreeCAD
- Python
- Voice engine dependencies

Network

Optional depending on the ASR implementation.

3. System Functions

Section	Functionality
5.1	Voice capture and feedback
5.2	primitive creations
5.3	dimensional parameters
5.4	Appearance and visual properties
5.5	Document management and security
5.6	Accessibility module and support

4. External Interface Requirements

4.1 User Interfaces

The system shall include an integrated or auxiliary DAV panel displaying:

- Microphone status
- Recognized text
- Errors
- Suggestions
- Confirmations

It shall also include:

- Start/stop button
 - Voice confirmation
 - Accessibility settings
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4.2 Hardware Interfaces

- Microphone
 - Optional audio output
 - Keyboard and mouse
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4.3 Software Interfaces

- FreeCAD API
 - ASR/NLU engine
 - FCStd files
 - Local configuration
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5. System Functions / Modules

5.1 Voice Capture and Feedback

Priority:

Must

Functional Requirements

REQ-1.1

The system shall clearly indicate when the microphone is active.

REQ-1.2

The system shall stop audio capture when listening mode ends.

REQ-1.3

The system shall display the recognized text.

REQ-1.4

The system shall request confirmation when confidence is low.

5.2 Voice-Based Primitive Creation

Priority:

Must

REQ-2.1

Create box.

REQ-2.2

Create cylinder or sphere.

REQ-2.3

Automatically create a document if none exists.

5.3 Dimensional Parameters

- Priority: Must

REQ-3.1

The system shall update part parameters via explicit voice commands.

REQ-3.2

The system shall request clarification from the user in the presence of an ambiguous command.

REQ-3.3

The system shall reject invalid or out-of-range numerical values.

5.4 Appearance

- Priority: Should

REQ 4.1

The system shall allow changing the color of the selected geometry via voice.

REQ-4.2

The system shall allow the assignment of basic materials via spoken commands.

5.5 Document and security

- Priority: Must/Should

REQ 5.1

The system shall allow creating, opening, and saving .FCStd files via voice commands.

REQ-5.2

The system shall request explicit confirmation (via voice or UI) before executing destructive actions, such as closing without saving.

5.6 Accessibility

- Priority: Must

REQ 6.1

The system shall offer alternative usage via keyboard and mouse in case of voice recognition failures.

REQ-6.2

The system shall emit clear sound or visual cues during interpretation errors to guide users with cognitive or visual difficulties.

6. Non-Functional Requirements

6.1 Performance

NF-1.1

The system shall respond within acceptable time limits.

NF-1.2

The system shall not block FreeCAD.

6.2 Security

NF-2.1

Communications shall use TLS.

NF-2.2

Credentials shall not be stored in plain text.

NF-2.3

Audio shall not be stored without consent.

6.3 Usability

NF-3.1

Help shall be available.

NF-3.2

Errors shall suggest corrective actions.

6.4 Reliability

NF-4.1

The system shall not corrupt files.

NF-4.2

Commands shall be versioned.

6.5 Portability

NF-5.1

Differences between operating systems shall be documented.